Name: _____

Class: _____Date:_____

CHAPTER II REVIEW

A. MULTIPLE-CHOICE QUESTIONS

Choose the correct answer.

1. Which statement below is **false**?

(A) A number divisible by 6 is always divisible by 2;
(B) If two numbers are both not divisible by 6, then their sum is not divisible by 6;
(C) If two numbers are both divisible by 6, then their sum is divisible by 6;
(D) A number ending in 0 is always divisible by 5.
Answer: _____

2. Which of the following numbers is a prime number?

(A) 3 045; (B) 2 139; (C) 523; (D) 4 718.
Answer: _____

3. Which of the following numbers is **not** a prime number?

(A) 31; (B) 4 115; (C) 89; (D) 433.
Answer: _____

4. Which of the following numbers is divisible by 9?

(A) 4 731; (B) 5 643; (C) 2 386; (D) 7 214.
Answer: _____

5. Which of the following numbers is divisible by 9 but not divisible by 5?

(A) 45 630; (B) 62 145; (C) 36 261; (D) 52 314.

Answer: _____

6. Which statement below is **false**?

(A) If a is not divisible by c and b is not divisible by c, then LCM(a, b) is not divisible by c;

(B) A common divisor of two natural numbers a and b is a divisor of their greatest common divisor;

(C) A common multiple of two natural numbers a and b is a multiple of their least common multiple;

(D) GCD(a, b) is a divisor of LCM(a, b).

Answer: _____

B. EXERCISES

2.56. Is each sum below a prime number or a composite number?

a) $3 \cdot 11 \cdot 14 + 55 \cdot 61$; b) $5 \cdot 6 \cdot 7 + 2\,024 \cdot 2\,025 \cdot 2\,026$.

_____

_____

2.57. Calculate, then write the result as a product of prime factors:

a) $15^2 : 9 + 3 \cdot 5$; b) $4 \cdot 5^2 - 48 : 2^3$.

_____

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2.58. The number of Grade 6 students in a school is between 300 and 400. When they line up in rows of 12, rows of 15 or rows of 18, there are always 7 students left over. Find the number of Grade 6 students.

2.59. Given A = 41 230 + 23 115 + 630. Without calculating, decide whether A is:

a) divisible by 2?

b) divisible by 5?

c) divisible by 3?

d) divisible by 9?

2.60. Two numbers have LCM $2^4 \cdot 3^3 \cdot 7^2$ and GCD $2^2 \cdot 7$. One of the numbers is $2^4 \cdot 7$. Find the other number.

2.61. If we multiply 15 873 by any non-zero one-digit number a and then multiply the result by 7, we get a 6-digit number whose digits are all a. For example, when a = 4:

15 873 · 4 = 63 492; 63 492 · 7 = 444 444.

Explain why.

2.62. Find the natural numbers n such that $10 \vdots (n + 2)$.

2.63. The two numbers $2^a \cdot 5^4$ and $2^5 \cdot 5^b$ have GCD $2^3 \cdot 5^2$ and LCM $2^5 \cdot 5^4$. Find the natural numbers a and b.

2.64. Calculate:

a) $\frac{5}{12} + \frac{7}{18}$;

b) $\frac{11}{15} - \frac{3}{10}$.